

22 Quantum physics

A · 2026 examination · Concept guide and original exam-style practice

Understand the concept

A photon has energy hf and momentum h/λ . Photoelectric emission requires photon frequency above a material's threshold: maximum kinetic energy = $hf - \text{work function}$. Raising intensity increases emitted electron count at fixed frequency above threshold, while raising frequency increases maximum kinetic energy.

Worked example: follow the reasoning

If photon energy is 4.0 eV and work function is 2.5 eV, maximum photoelectron kinetic energy is 1.5 eV.

Quick check · multiple choice

A What increases the maximum kinetic energy of photoelectrons above threshold? [1]

- A Light intensity alone
- B Light frequency
- C Illuminated area alone
- D Exposure time alone

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A metal's threshold frequency is 5.0×10^{14} Hz. Light of 8.0×10^{14} Hz falls on it. [3]
Find maximum kinetic energy, $h = 6.63 \times 10^{-34}$ J s.

Final answer: _____

Answers and explanation

Quick check: B. Light frequency

Question 1 [3 marks]

Use $K_{\text{max}} = h(f - f_0)$ [1]; frequency difference 3.0×10^{14} Hz [1]; $K_{\text{max}} = 1.99 \times 10^{-19}$ J [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Physics 9702, 2025–2027. <https://www.cambridgeinternational.org/Images/664565-2025-2027-syllabus.pdf>

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